

Automated Customer Reviews Analysis

Objectives:

- Sentiment Classification of Customer Reviews
- Clustering Product Categories
- Identifying Top & Worst Products with Key Complaints

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Task 1 – Sentiment Classification

Preprocessing

Used Columns: reviews.text, reviews.rating

Sentiment Mapping:

```
def map_sentiment(rating):  
    if rating in [1, 2]:  
        return 'negative'  
    elif rating == 3:  
        return 'neutral'  
    else:  
        return 'positive'
```

Task 1 – Sentiment Classification

Preprocessing :

Class Distribution Before Balancing:

- Positive: 62,579
- Neutral: 2,902
- Negative: 2,510

Class Distribution After Balancing:

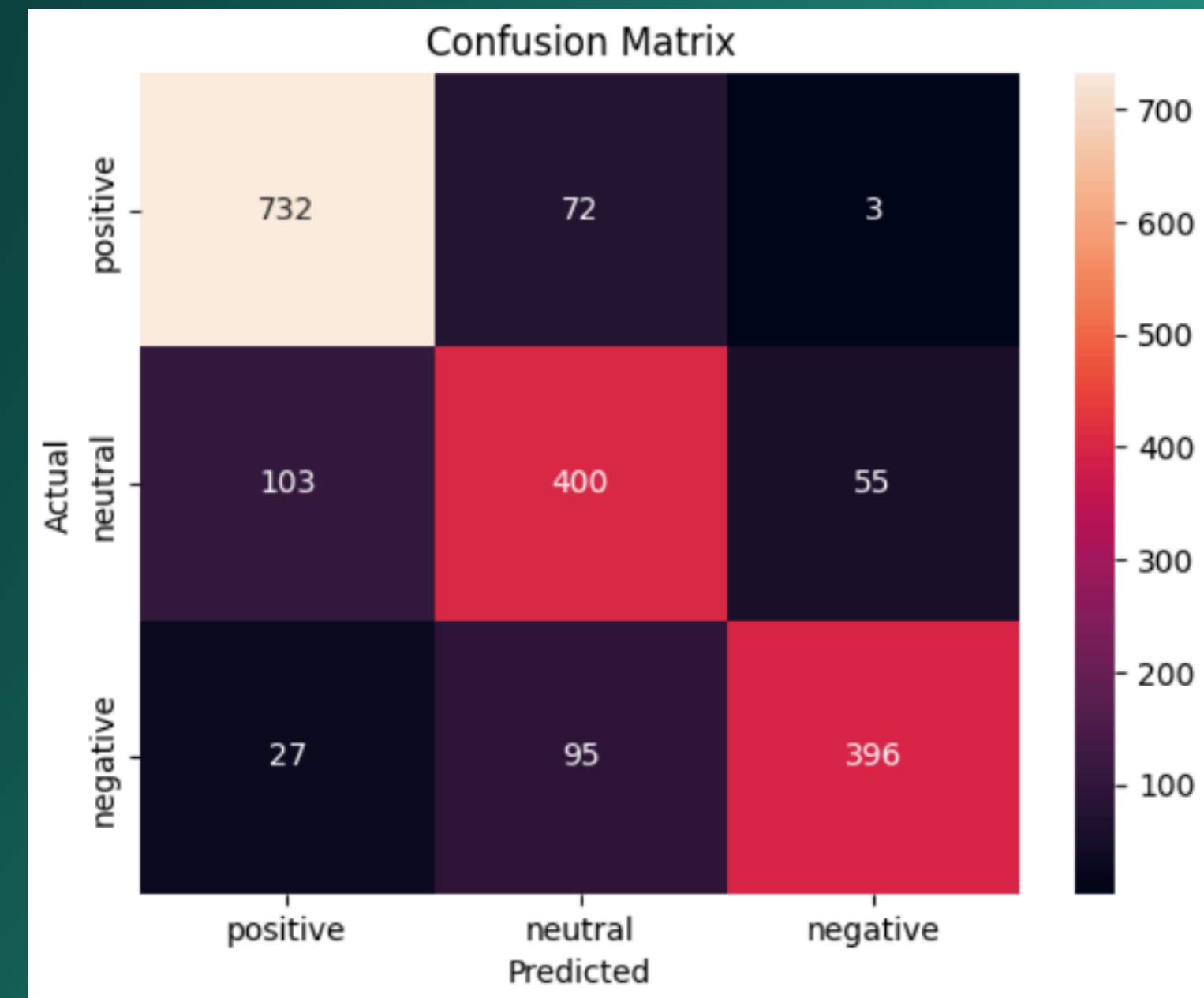
- Positive: 4,000
- Neutral: 2,902
- Negative: 2,510

Task 1 – Sentiment Classification

Model :
DistilBertForSequenceClassification

Evaluation Results :

- Accuracy: 81.15%
- Classification Report:
 - Positive: Precision=0.85, Recall=0.91, F1=0.88
 - Neutral: Precision=0.71, Recall=0.72, F1=0.71
 - Negative: Precision=0.87, Recall=0.76, F1=0.81

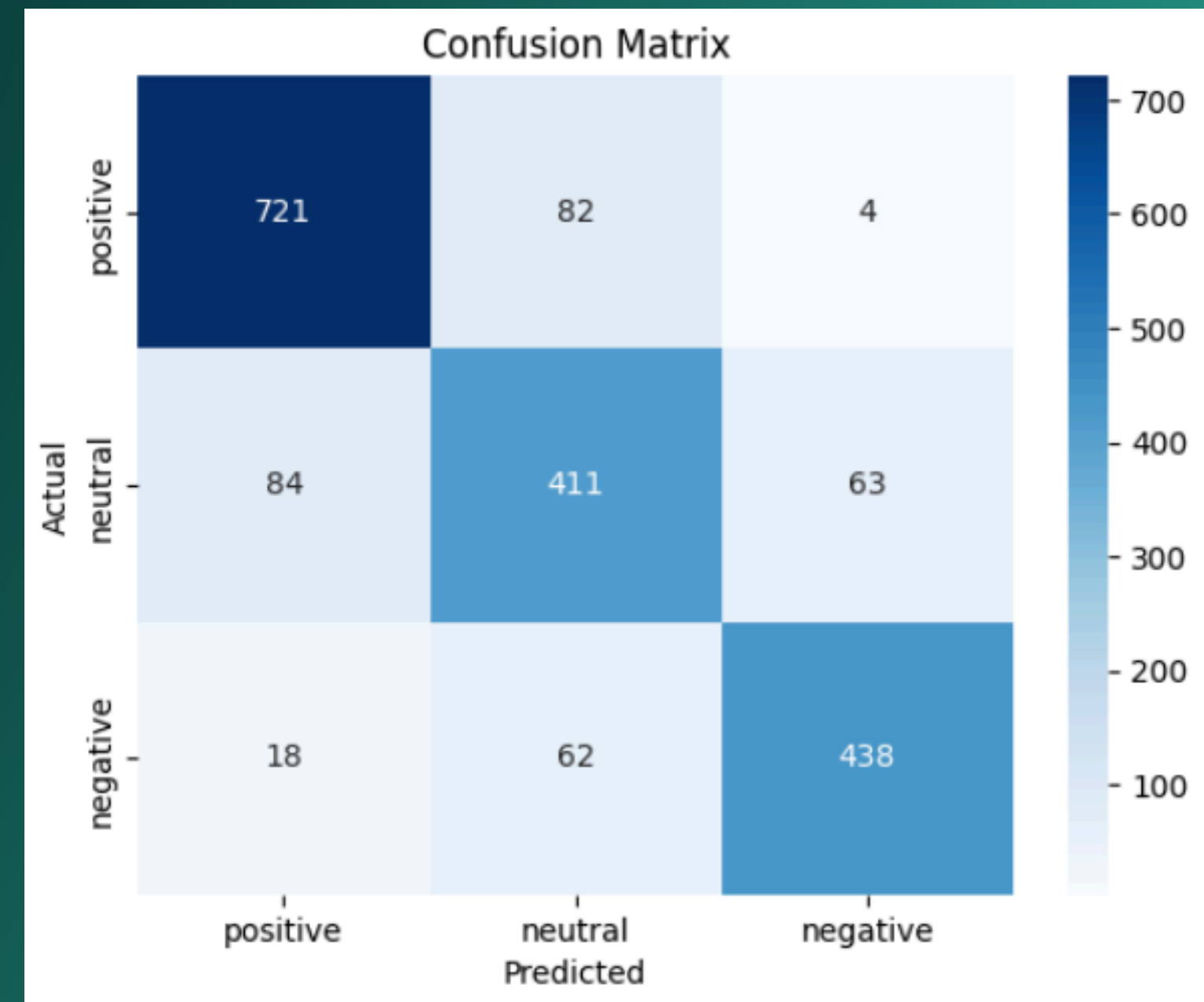


Task 1 – Sentiment Classification

Model :
RobertaForSequenceClassification

Evaluation Results :

- Accuracy: 83.38%
- Classification Report:
 - Positive: Precision=0.88, Recall=0.89, F1=0.88
 - Neutral: Precision=0.74, Recall=0.74, F1=0.74
 - Negative: Precision=0.87, Recall=0.85, F1=0.86



Task 2 – Product Category Clustering

Used Columns: name (unique values only)

Total Unique Product Names: 79

Feature Extraction: TF-IDF Vectorization

Clustering Algorithm: KMeans (5 clusters)

Cluster Labels Mapping:

```
cluster_name_map = {  
    0: 'AmazonBasics Essentials',  
    1: 'Kindle E-Readers',  
    2: 'Alexa Devices & Accessories',  
    3: 'Fire Tablets',  
    4: 'Pet Supplies & Laptop Backpacks'  
}
```

Task 3 – Product Ratings & Complaints Analysis

--- 📁 Alexa Devices & Accessories ---

If you're in the market for some new Alexa devices and accessories, look no further! We've rounded up the top 3 products

1. ****Certified Refurbished Amazon Echo**:**

- This smart speaker is a fan favorite for its voice-controlled capabilities and seamless integration with Alexa.
- Key Differences: The Certified Refurbished version offers the same great features at a more affordable price point.
- Top Complaints: Users have not reported significant complaints about this product.

2. ****Fire TV Stick Streaming Media Player Pair Kit**:**

- With this pair kit, you can stream your favorite shows and movies in high definition on any TV.
- Key Differences: The pair kit includes two Fire TV Stick devices, perfect for multiple TVs in your home.
- Top Complaints: Users have not reported significant complaints about this product.

3. ****Expanding Accordion File Folder Plastic Portable Document Organizer Letter Size**:**

- Stay organized with this portable file folder that can hold all your important documents and papers.
- Key Differences: This accessory is a must-have for keeping your paperwork in order and easily accessible.
- Top Complaints: Users have not reported significant complaints about this product.

Task 3 – Product Ratings & Complaints Analysis

--- 📁 Fire Tablets ---

Looking for the perfect Fire Tablet to suit your needs? Look no further! We've compiled a list of the top 3 Fire Tablets on the market.

1. ****All-New Fire HD 8 Tablet, 8" HD Display, Wi-Fi, 32 GB - Includes Special Offers, Magenta****: This sleek and powerful tablet is perfect for entertainment and productivity.

2. ****All-New Fire HD 8 Kids Edition Tablet, 8 HD Display, 32 GB, Pink/Blue Kid-Proof Case****: Designed specifically for kids, these tablets offer a safe and fun experience.

****Key Differences****: The main difference between the Kids Edition and the regular Fire HD 8 Tablet is the added durability of the kid-proof case.

****Top Complaints****:

- ***All-New Fire HD 8 Tablet***: No significant complaints reported.
- ***All-New Fire HD 8 Kids Edition Tablet (Pink)***: Ads, early shutdown issues, and confidence-reducing product failures were the top complaints.
- ***All-New Fire HD 8 Kids Edition Tablet (Blue)***: Shutdown issues, defective products, and cumbersome parental controls were the main concerns.

****Worst Product to Avoid****: Unfortunately, the Fire Tablet, 7 Display, Wi-Fi, 16 GB - Includes Special Offers, Black, has received a lot of negative feedback.

In conclusion, when choosing a Fire Tablet, consider your specific needs and preferences to select the best option for you. Whether it's for entertainment or productivity, there's a Fire Tablet for everyone.

Task 3 – Product Ratings & Complaints Analysis

--- 📁 Kindle E-Readers ---

If you're an avid reader looking for the perfect E-reader, look no further than the Kindle E-readers lineup. Let's dive into the top 3

- **1. Kindle Voyage E-reader, 6 High-Resolution Display (300 ppi) with Adaptive Built-in Light, PagePress Sensors, Free 3G + Wi-Fi - Includes Special Offers*
- **2. Kindle Oasis E-reader with Leather Charging Cover - Black, 6" High-Resolution Display (300 ppi), Wi-Fi - Includes Special Offers*
- **3. Kindle Voyage E-reader, 6 High-Resolution Display (300 ppi) with Adaptive Built-in Light, PagePress Sensors, Wi-Fi - Includes Special Offers*

Key Differences:

- The Kindle Voyage with Free 3G offers additional connectivity options compared to the Wi-Fi only version.
- The Kindle Oasis comes with a luxurious leather charging cover for added style and convenience.
- Both the Voyage models feature Adaptive Built-in Light and PagePress Sensors for a premium reading experience.

User Feedback:

- The Kindle Oasis received complaints about weak magnetic connectors and a poorly designed charging setup.
- Some users of the Wi-Fi only Voyage model reported issues with software updates causing device malfunctions, leading to frustration and returns.
- Despite some negative feedback, the Kindle Voyage with Free 3G garnered positive remarks for its product quality and customer service.

Worst Product to Avoid:

**All-New Kindle Oasis E-reader - 7 High-Resolution Display (300 ppi), Waterproof, Built-In Audible, 32 GB, Wi-Fi - Includes Special Offers*

Based on user complaints, this Kindle Oasis model has faced criticism for various issues, although specific complaints were not provided.

In conclusion, when choosing a Kindle E-reader, consider your connectivity needs, design preferences, and user feedback to find the perfect device for your reading habits.

Task 3 – Product Ratings & Complaints Analysis

--- 📁 AmazonBasics Essentials ---

Looking for reliable and affordable tech accessories and storage solutions? Look no further than the AmazonBasics Essentials range! In this a

1. **Top 3 Recommended Products:**

- **AmazonBasics USB 3.0 Cable - A-Male to B-Male - 6 Feet (1.8 Meters):** This high-speed USB cable is perfect for connecting your printer,
- **AmazonBasics 15.6-Inch Laptop and Tablet Bag:** Keep your laptop and tablet safe and secure with this padded bag. It's designed to fit de
- **AmazonBasics External Hard Drive Case:** Protect your portable hard drive with this durable case. The interior strap and mesh pocket keep

2. **Key Differences:**

- The USB cable is essential for connecting devices, the laptop bag is perfect for on-the-go protection, and the hard drive case ensures t

3. **Top Complaints:**

- *AmazonBasics 15.6-Inch Laptop and Tablet Bag:* Some users mentioned that the bag was of lower quality than expected and was a tight fit

4. **Product to Avoid:**

- **AmazonBasics 11.6-Inch Laptop Sleeve:** Based on user complaints, this laptop sleeve may not offer the quality or fit that users desir

In conclusion, the AmazonBasics Essentials range offers a variety of affordable and practical products to enhance your tech experience. Wheth

Challenges & Solutions

Class Imbalance in Sentiment Labels :

- Challenge: Highly skewed data with a majority of positive reviews.
- Solution: Downsampled the positive class to balance the dataset before training.

Noisy or Duplicate Product Names :

- Challenge: Inconsistent product naming affected clustering accuracy.
- Solution: Applied preprocessing and selected unique product names for clustering.

**Thanks for
Listening!..**